United International University
Department of Computer Science and Engineering

CSE 3421: Software Engineering Final: Summer 2025

Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. The numbers on the right of the questions denote their marks.

1. Design **UML diagrams** using the **appropriate design pattern** for the following scenarios: [CO2]
   (a) SkyByte manufactures drones for three models: Falcon, Hawk, and Eagle. Each model has a distinct Camera System and Propulsion Unit. Falcon uses a 4K UltraCam with a TurboJet Propulsion, Hawk uses a ThermalCam with a QuadCopter Propulsion, and Eagle uses a NightVisionCam with a Hybrid Rotor Propulsion. (4)

   (b) SkyByte allows users to enhance drones with add-ons like Auto-Return System, Obstacle Avoidance, and Live Tracking Module. These add-ons can be attached independently to any drone. (4)

2. (a) RideGo, a ride-sharing app overwhelms users by placing too many options on a single booking screen like vehicle type, coupons, ride preferences, wallet balance, map options, promotions, and fare details resulting in a cluttered layout. Meanwhile, the most important action, Confirm Ride, is buried near the bottom and often overlooked. The app provides no onboarding hints or tooltips for new users and has no shortcuts or personalization for frequent users. As a result, many new customers abandon rides halfway out of frustration. Which of the **8 Schneiderman's Golden Rules** of Interface Design are **violated** in the above scenario? Justify with **reasoning** and **suggest design changes** to remedy the violations. [CO1] (5)

   (b) **Explain** what is meant by **Regression testing** and **Smoke testing**. Find the **differences** between these two types of testing. [CO1] (3)

3. (a) A startup named HomeEase is designing a Smart Apartment Control System that connects devices such as door sensors, motion detectors, temperature monitors, indoor lights, alarm units, and security cameras. These devices occasionally send notifications to the system and based on the situation, different automated actions must be triggered. For example, turning on lights, sounding an alarm, or starting a short camera recording. The system must allow these automated reactions to run independently so if one action fails or becomes unresponsive, the others can continue without interruption. The architecture should also support easy addition of new automated reactions in the future without changing existing components. Which **architectural pattern** will be best suited for the software above? Explain with **proper reasons**. Also **explain** how your selected architecture **functions**. [CO2] (5)

   (b) **Compare** the **API** protocols **REST** and **SOAP**. [CO2] (3)

4. (a) Raymond Reddington, a recent CS graduate, had very limited knowledge of Java programming. However, he claimed to be an expert in Java on his CV and secured a position that required strong Java skill. His employer, unaware of his inexperience, assigned him to develop a Java-based hospital management system. Instead of improving his skills or consulting colleagues, Raymond chose to depend entirely on AI-generated code, copying it directly into the project without testing or validation. The resulting software contained several bugs that compromised the reliability of patient data management in the hospital. Which of the **ACM principles of ethics** are violated in the above scenario? Explain with **proper reasoning**. [CO3] (5)

   (b) Explain **three** key characteristics of **responsible and ethical AI** in the context of software engineering. [CO2] (3)

5. (a) Find the **Earliest Start, Earliest Finish, Latest Start, Latest Finish, Project Deadline, and Critical Path** using the **Critical Path Method** for the following tasks: [CO2] (8)

Tasks	Duration	Precedents
A	8	B, C
B	4	-
C	10	-
D	5	A
E	6	D
F	8	-
G	7	F
H	7	A
I	5	H
J	8	D, G
K	9	H, J